



Basic operations

We can remotely log in to a machine by issuing the following command:

```
slymox@gnubox~$ ssh user@hostname
```

Here, 'user' is an existing user on the remote machine 'hostname', so you need to replace the two with relevant information. [You can also use an IP address instead of a hostname to log in.] Hitting the **Enter** key now will result in a prompt for the user's password; and after entering it, you will get the remote user's shell prompt.

Alternately, we can also provide the following command:

```
slymox@gnubox~$ ssh hostname
```

...which is equivalent to:

```
slymox@gnubox~$ ssh slymox@hostname
```

...i.e., if the user name of the one trying to remotely log in is the same as the current user, there is no need to specify the user name explicitly.

Sometimes systems administrators will configure the SSH daemon to listen to a non-standard port such as 422, instead of 22. This is done for security reasons—to make it difficult for an unauthorised person to easily find which port number the SSH daemon is listening to.

In cases where we need to perform the SSH log-in via a non-standard port, we can specify the port number explicitly using the *-p* option:

```
slymox@gnubox~$ ssh -p 422 slymox@hostname
```

The initial key discovery

When you connect to an SSH server for the first time, you will be asked to verify the server's key. When the users continue confirming 'yes', it will attach the server key with the hostname and store it in the *~/.ssh/known_hosts* file. After the initial probe for the server verification, it will check this *known_hosts* file to verify the authority of the server to which the SSH client is requesting a connection to.

```
slymox@gnubox~$ ssh slymox@192.168.1.2
The authenticity of host '192.168.1.2 (192.168.1.2)' can't be established.
RSA key fingerprint is 6d:92:2c:f1:74:e7:a9:21:64:57:90:6f:72:3e:a3:18.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.168.1.2' (RSA) to the list of known hosts.
slymox@192.168.1.2's password:
Last login: Sun May 17 21:04:29 2009 from slymox-laptop
slymox@gnubox~$
```

This initial key discovery process is to ensure security. It is possible for an attacker to steal information from the remote user log-in by impersonating the server, i.e., if the attacker can provide a server with the same host name and user authentication, the user connecting from the remote machine will be logged into a fraud machine and data may be stolen.

Each server will have a randomly generated RSA server key.

To ensure security, in cases where the server key changes, the SSH client will issue a serious warning reporting that the host identification has failed and that it will stop the log-in process.

```
slymox@gnubox~$ ssh slymox@192.168.1.2
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@@@@@@@@@@@@@@@@@@@@
@ WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED! @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that the RSA host key has just been changed.
The fingerprint for the RSA key sent by the remote host is
```